Baijiang (Noke) Yuan

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Education

University of Toronto, PhD in AI in Healthcare · GPA: 4.0/4.0 09/2022-present
Advisors: Prof. Geoffrey Liu, Dr. Robert Grant
Research: Developing ML systems for real-time cancer symptom prediction and clinical decision support.
University College London, UK, MSc in Health Data Science · GPA: 3.3/4.0 09/2020-09/2021
University of Toronto, HBSc in Computer Science Specialist · Statistics Major · GPA: 3.3/4.0 09/2015-11/2019

Research interests

Machine learning in cancer research · Computational genomics · Medical imaging · Large language models in healthcare

Industrial experience

Machine Learning Associate, Vector Institute, Toronto, ON, Canada 04/2026-10/2026

- *Built a full-stack computer vision system (FastAPI backend, React frontend) for Deighton Associates' dTECT platform, automating roadway asset inventory and pavement-condition assessment from dash-cam imagery.*
- *Trained and benchmarked detection models for pavement type, distress severity, and roadway assets; explored vision-language models (VLMs) for zero-shot detection, delivered via an interactive dashboard.*

Machine Learning Engineer Intern, UHN AI Hub, Toronto, ON, Canada 05/2025-10/2025

- *Developed Clinical-Camel-ASR, an end-to-end medical scribing framework converting doctor–patient conversations into structured clinical summaries using ASR and LLMs.*
- *Collaborated with WangLab (Dr. Bo Wang @ UofT) on healthcare NLP, supervised by Dr. Jun Ma and Alif Munim.*

Data Analyst Intern, NetEase, Beijing, China 04/2020-09/2020

- *Built statistical risk models analyzing 1M+ user records for supplier risk management and strategic decision-making.*
- *Conducted large-scale data analysis to identify trends, assess risks, and optimize supplier interactions.*

Research experience

Research Trainee, Data Sciences Institute, University of Toronto, Toronto, ON, Canada 04/2025-07/2025

- *Participated in the Galvanizing Data Science Applications in Early Stage Drug Discovery program hosted by the Data Sciences Institute and Structural Genomics Consortium.*
- *Received hands-on training in machine learning for hit-finding, experimental screening data generation, and collaborative drug discovery workflows.*

Research Student, Princess Margaret Cancer Centre, UHN, Toronto, ON, Canada 05/2022-present

- *First-author of ML symptom prediction system (JCO CCI 2025), validated across 82 cancer centers (12K+ patients, 77K+ treatments), achieving AUROC 0.66–0.73 for 9 symptoms.*
- *Developed AIM2REDUCE real-time prediction system for 30-day symptom deterioration, with high-risk alerts associated with 1.85–2.66× increased ED visits.*

Machine Learning Intern, UCL Institute of Health Informatics, London, UK 01/2022-05/2022

- *Developed multi-task deep learning system using YOLO for automated detection of Age-Related Macular Degeneration from retinal fundus images.*
- *Built CNN-based COVID-19 detection pipeline from lung MRI and EHR-based prognostic models for clinical outcomes.*

Research Student, Human Genetics Lab, University College London, London, UK 01/2021-08/2021

- *Validated plasma volume status as a heart failure biomarker using UK Biobank data (6,982 cases, 327,455 controls).*
- *Conducted GWAS identifying 438 significant genetic loci and applied Mendelian randomization for causal inference.*

Selected publications

1. **B. Yuan**, M. Kabir, J.C. He, Y. Li, B. Grant, S. Narine, M. Welch, S. Podolsky, N. Liu, R. Ajaj, L.J. Zhan, et al., "Development of Machine Learning Systems to Predict Cancer-Related Symptoms With Validation Across a Health Care System." *JCO Clinical Cancer Informatics*, 9:e2500073, 2025.
2. W.I.T. Uy, G.A.G. Conchas, J.C. He, M. Kabir, **B. Yuan**, G. Liu, S. Narine, M. Powis, B. Grant, M. Welch, T. Truong & R. Grant, "Prompting Large Language Models to Predict Adverse Events during Cancer Treatment." *Clinical Cancer Research*, 31(13_Supplement):B012, 2025.
3. J. Siraj, M. Kabir, S. Kim, J.C. He, **B. Yuan**, W. Uy, T. Patel, B. Grant, S. Narine, M. Krzyzanowska, T. Truong, G. Liu, C. McElcheran, R. Grant & M. Welch, "Jarvais: A Modular Framework to Standardize Machine Learning Workflows and Accelerate Reproducible AI in Oncology." *Clinical Cancer Research*, 31(13_Supplement):B001, 2025.
4. J.C. He, H. Alghamni, I. Hirsch, **B. Yuan**, M.M. Kabir, G. Liu, M.L. Powis, E. Yeo, P. Gross, B. Grant, S. Narine, M. Welch, T. Truong & R.C. Grant, "Dynamic Prediction of Cancer-Associated Thrombosis to Guide Prophylactic Anticoagulation." *Journal of Clinical Oncology*, 43(16_suppl):e13696, 2025.
5. J.C. He, I. Hirsch, Y. Li, **B. Yuan**, M. Kabir, B. Grant, S. Narine, M. Welch, W. Xu, M.K. Krzyzanowska, M.L. Powis, G. Liu, T. Truong & R.C. Grant, "Impact of Applying Machine Learning to the Electronic Medical Record on Prediction of Cancer-Associated Thrombosis." *JCO Oncology Practice*, 20(10_suppl):409, 2024.
6. M. Kabir, J.C. He, **B. Yuan**, V. Tran, K. Sue, T. Patel, B. Grant, Y. Li, S. Narine, M. Welch, M.K. Krzyzanowska, T. Truong, G. Liu & R.C. Grant, "Prospective Silent Deployment and Evaluation of an Intelligent Machine Learning Model for Prediction of Emergency Department Visits during Cancer Treatment." *JCO Oncology Practice*, 20(10_suppl):407, 2024.
7. **B. Yuan**, A. Schmid, S. Yang, A. Hammad, D. Gawish, M. Gresham, K. Zhao, R. Shah, G. Bradbury, J. Mendes De Carvalho, J. Leighl, R.C. Grant & G. Liu, "Multi-Task Machine Learning of Electronic Medical Records to Predict Future Symptoms in Cancer Patients." *JCO Oncology Practice*, 19(11_suppl):587, 2023.
8. R.C. Grant, **B. Yuan**, A. Gawish, A. Schmid, A. Hammad, E. Chen, J. Wang, D. Jani, K. Eng, M. Gresham, S. Yang, K. Zhao, R. Shah, G. Bradbury, M. Rodriguez, J.M. De Carvalho, J. Leighl & G. Liu, "AIM2REDUCE: Real-Time Early Detection of Undesirable Cancer Events." *Journal of Clinical Oncology*, 41(16_suppl):1557, 2023.

Selected conference presentations

1. **B. Yuan**, *Multi-task Machine Learning of Electronic Medical Records Predicts Future Symptoms in Cancer Patients*, ARCC Annual Conference, Ottawa, Canada. 2024.
2. **B. Yuan**, *Multi-task Machine Learning of Electronic Medical Records Predicts Future Symptoms in Cancer Patients*, T-CAIREM AI in Medicine Conference, Toronto, Canada. Nov 2023.
3. **B. Yuan**, *Multi-task Machine Learning of Electronic Medical Records Predicts Future Symptoms in Cancer Patients*, ASCO Annual Meeting Symposium, Boston, MA, USA. Jun 2023.

Selected honors and awards

Google x T-CAIREM Datathon, First Place Winner 2025
Institute of Medical Science Open Fellowship Award ($15,000), University of Toronto 2023-2025

Skills

Programming: Python (TensorFlow, PyTorch, scikit-learn, pandas, NumPy), R, SQL, Git, Linux/Bash
Technical: Machine learning, deep learning, large language models, computational genomics, bioinformatics, medical imaging, statistical analysis
Languages: English (Full Professional), Mandarin (Native), Cantonese (Full Professional)

References

Dr. Geoffrey Liu
Professor, Departments of Medicine, Medical Biophysics, Pharmacology, Toxicology, Epidemiology
University of Toronto
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Dr. Robert Grant
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